Bablu Kumar Patel

Roll No.: 25CEM3R06
Master of Technology
Transportation Engineering
National Institute Of Technology, Warangal

+91-6306629410
bkp0907@gmail.com
bk25cem3r06@student.nitw.ac.in
Portfolio
LinkedIn Profile

Education

•National Institute Of Technology, Warangal *2025 – 2027*
Master of Technology in Transportation Engineering CGPA: 7.38

•Rajkiya Engineering College, Azamgarh *2021 – 2025*
Bachelor of Technology in Civil Engineering CGPA: 7.25

•Ratna Memorial Public School *2021*
Board of Senior Secondary Education, Uttar Pradesh Percentage: 92.6%

Work Experience

•Public Works Department (PWD) *July 2024 – Aug 2024*
Summer Intern Gorakhpur, Uttar Pradesh

- Completed a 4-week intensive field exposure program on public infrastructure projects, focusing on road and pavement works.
- Assisted in surveying operations including differential leveling, traversing, and preparation of cross-sections for road alignment.
- Supported quality control of construction materials through soil sample collection, laboratory testing, and pavement layer inspection.

Personal Projects

•ML-Based Prediction of Marshall Stability of Waste Modified Asphalt Concrete *2024 – 2025*
B.Tech Final Year Project on sustainable materials in asphalt concrete

- Prepared asphalt mixes with SCBA as filler at 0%, 1%, 3% and 5% replacement levels and reclaimed aggregates as coarse aggregate; varied bitumen content from 4.0% to 8.0%.
- Cast and tested multiple Marshall specimens in accordance with IS and ASTM standards to evaluate Marshall Stability, flow value, air voids and volumetric properties.
- Performed Support Vector Machine (SVM) based modeling on experimental mix design data achieving 99% training accuracy (R^2=0.99).
- Demonstrated the potential of SCBA and reclaimed aggregates to improve sustainability of flexible pavements while maintaining acceptable performance.

•Trip Generation Modeling: Neural Networks vs Linear Regression *2025*
Comparative analysis of ML techniques for transportation planning

- Developed MLP neural network for trip generation modeling achieving R^2 of 94.2%, outperforming traditional linear regression by 4.8x.
- Processed 1,800+ household records using feature engineering and data preprocessing techniques.
- Identified earning members and household size as primary predictors through variable importance analysis.

•Safety Analysis of a Median Opening on a Horizontal Curve at NIT Warangal 2nd Gate *2025 – 2026*
Minor Project – Transportation Engineering, NIT Warangal

- Conducted a field-based traffic safety study on NH-163 at NIT Warangal 2nd Gate, analyzing the combined effect of horizontal curvature, median opening geometry, and mixed traffic behavior on road safety.
- Analyzed Stopping Sight Distance and gap acceptance behavior on NH-163, identifying critical geometric inadequacies where available SSD (25m) fell significantly below required threshold, causing high collision risk.
- Quantified forced merging behavior under mixed traffic – 38.6% of two-wheelers and 25% of cars merged unsafely below critical gap thresholds of 2.5s and 3.5s respectively.
- Proposed geometric and traffic management improvements to reduce conflict points at a complex median opening on a national highway.

•**Advanced Travel Demand Modelling: Activity-Based Study at NIT Warangal** *2025 – 2026*

Comparative analysis of mode choice and activity patterns using discrete choice models

- Conducted an activity-based travel demand study on 144 NITW campus respondents using structured activity diaries, analysing trip chaining behaviour across 285 person-day chains; Complex Work/Study chains were the most dominant pattern (35.79%), with complex chains constituting 81.4% of all observed chains.
- Quantified mode share distribution with walking (56.9%) and cycling (22.9%) accounting for ∼80% of all trips; identified residential proximity as the primary determinant of mode choice through gender and residence-type disaggregated analysis.
- Estimated and compared Nested Logit, Full MNL, and Restricted MNL models for mode choice prediction; Nested Logit was found unsuitable ($\mu > 1$) and Restricted MNL was identified as the best-fit model (AIC: 2043.86, Hit Rate: 62.69%).
- Proposed evidence-based policy recommendations including enhanced pedestrian infrastructure and campus cycling facilities to promote sustainable non-motorised mobility on campus.

Technical Skills and Interests

Software: PTV VISSIM, PTV VISUM, SIDRA INTERSECTION, Civil 3D, IIT PAVE, STAAD.Pro, AutoCAD (2D Drafting), AIMSUN
Data Analysis & Modelling: IBM SPSS, SVM Modelling, Excel, Python (NumPy, Pandas, Matplotlib)
Laboratory & Materials Testing: Marshall Stability Testing, Asphalt Mix Design, Aggregate Testing, Concrete Testing, Soil Mechanics (CBR, Compaction, Atterberg Limits)
Areas of Interest: Transportation Engineering, Sustainable Infrastructure, Machine Learning Applications in Civil Engineering
Languages: English (Fluent), Hindi (Native)

Positions of Responsibility

•**Media Head,** Innoventure's Club, Rajkiya Engineering College Azamgarh *6 Months*

•**Fest Coordinator,** College Tech Fest, Rajkiya Engineering College Azamgarh

Certifications & Workshops

•**AutoCAD Training** – Internshala (8-Week Training, Top Performer) *2024*

•**STAAD Pro Training** – Internshala (8-Week Training) *2024*

•**Developing Soft Skills and Personality** – NPTEL

•**Enhancing Soft Skills and Personality** – NPTEL

•**AutoCAD 2D Drafting and Design Workshop** – Learn Delta

Achievements

- **GATE 2025 Qualified (Civil Engineering)** Successfully qualified in the national-level Graduate Aptitude Test in Engineering.